SIMULATION EXPERIMENT – 13

● Simulation-Based Frequency-Domain Characterization of a Microwave Two-Port Network Using S-Parameters and Network-Analyzer Emulation

➔ CO Mapping: CO4

Bloom's Level: K5 – Evaluate

WK Level: WK5 – VNA-oriented network analysis

➔ Objectives of this Experiment:

1. To develop a frequency-domain model of a microwave two-port network.
2. To generate simulated S11, S12, S21 and S22 responses over a specified frequency range.
3. To extract return loss, insertion loss, isolation and transmission characteristics from the simulated data.
4. To emulate the interpretation of VNA measurements and identify frequency regions of optimum network performance.

➔ Expected Outcome from this Experiment:

Students shall be able to:

1. Develop and simulate a frequency-domain model of a microwave two-port network.

2. Generate and analyze the simulated S11, S12, S21 and S22 responses over the specified frequency range.
3. Extract and evaluate return loss, insertion loss, isolation and transmission characteristics from the simulated S-parameter data.
4. Interpret and validate the simulated S-parameter responses in a VNA-like manner and identify the frequency regions corresponding to optimum network performance.

➔ Software Required: MATLAB / ADS / CST / HFSS

➔ Objective linkage: Develop a two-port frequency-domain model, generate S11/S12/S21/S22,

extract network performance parameters, and identify optimum frequency regions.

➔ Experimental Procedure:

A. VNA-Style S-Parameter Simulation: Simulate a 50 Ω microwave two-port network from 8 to 12

GHz using a frequency step of 10 MHz. Obtain S11, S12, S21 and S22. At the frequency where S21 is

maximum, calculate insertion loss and at the frequency where S11 is minimum calculate return loss.

Identify the operating region satisfying S11 ≤ −15 dB and S21 ≥ −3 dB.

B. Network Interpretation: Compare S11 and S22 to evaluate input/output matching and S12 to

evaluate reverse isolation. Determine the frequency interval over which the network simultaneously

satisfies return loss ≥15 dB and insertion loss ≤3 dB. Report the center frequency and bandwidth of

this acceptable region.

CODE:

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%%
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%  SIMULATION EXPERIMENT 13  (ECA13 Microwave Engineering, CO4, K5 / WK5)

%  Simulation-Based Frequency-Domain Characterization of a Microwave

%  Two-Port Network Using S-Parameters and VNA Emulation

%

%  DUT : 3rd-order Chebyshev band-pass two-port (coupled-resonator model,

%        finite unloaded Q so that real insertion loss is present)

%  Z0  : 50 ohm,  8-12 GHz,  step = 10 MHz

%
================================================================================
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clc; clear; close all;


%% 1. INPUT PARAMETERS (Table C)

fstart = 8e9;          % Start frequency  [Hz]
```

```
fstop  = 12e9;          % Stop frequency   [Hz]
df     = 10e6;          % Frequency step   [Hz]  (0.01 GHz)
Z0     = 50;            % Reference impedance [ohm]
f      = fstart:df:fstop;
N      = numel(f);

% Thresholds (Key Conditions)
S11_th = -15;           % S11 <= -15 dB  (RL >= 15 dB)
S21_th = -3;            % S21 >= -3 dB   (IL <= 3 dB)

%% 2. DUT DESIGN PARAMETERS (frequency-domain two-port model)
f0   = 10e9;            % Design centre frequency [Hz]
FBW  = 0.20;            % Fractional bandwidth (2 GHz)
n    = 3;               % Number of resonators
RLdes = 20;             % Designed passband return loss [dB]
Qu   = 300;             % Unloaded Q of resonators (gives finite IL)

% --- Chebyshev low-pass prototype g-values ---
LAr  = -10*log10(1 - 10^(-RLdes/10));       % passband ripple [dB]
beta = log(coth(LAr/17.37));
gam  = sinh(beta/(2*n));
a = zeros(1,n); b = zeros(1,n); g = zeros(1,n+2);
for k = 1:n
   a(k) = sin((2*k-1)*pi/(2*n));
   b(k) = gam^2 + sin(k*pi/n)^2;
end
g(1) = 1;                               % g0 (source)
g(2) = 2*a(1)/gam;                      % g1
for k = 2:n
```

```
    g(k+1) = 4*a(k-1)*a(k)/(b(k-1)*g(k));    % g2 ... gn
end
g(n+2) = 1;                              % g(n+1) (odd order -> 1)
gk = g(2:n+1);                           % g1..gn

% --- Coupling matrix & external Q (normalised) ---
m = zeros(n);
for i = 1:n-1
    m(i,i+1) = 1/sqrt(gk(i)*gk(i+1));
    m(i+1,i) = m(i,i+1);
end
qe1 = g(1)*gk(1);                        % input  external Q
qen = gk(n)*g(n+2);                      % output external Q
q   = zeros(n); q(1,1) = 1/qe1; q(n,n) = 1/qen;
delta = 1/(FBW*Qu);                      % loss term

%% 3. FREQUENCY-DOMAIN S-PARAMETER COMPUTATION
S11 = zeros(1,N); S21 = zeros(1,N);
S12 = zeros(1,N); S22 = zeros(1,N);
for idx = 1:N
    Om = (1/FBW)*(f(idx)/f0 - f0/f(idx));    % low-pass frequency variable
    A  = q + (delta + 1i*Om)*eye(n) - 1i*m;
    Ai = inv(A);
    S11(idx) = 1 - (2/qe1)*Ai(1,1);
    S22(idx) = 1 - (2/qen)*Ai(n,n);
    S21(idx) = -1i*2/sqrt(qe1*qen)*Ai(n,1);
    S12(idx) = -1i*2/sqrt(qe1*qen)*Ai(1,n);  % reciprocal network
end
```

```
% Magnitudes in dB
S11dB = 20*log10(abs(S11));
S21dB = 20*log10(abs(S21));
S12dB = 20*log10(abs(S12));
S22dB = 20*log10(abs(S22));

% Derived quantities  (Formulae section E)
RL  = -20*log10(abs(S11));      % Return loss  [dB]
RL2 = -20*log10(abs(S22));      % Output return loss [dB]
IL  = -20*log10(abs(S21));      % Insertion loss [dB]
ISO = -20*log10(abs(S12));      % Reverse isolation [dB]

%% 4. PART A : OPERATING POINTS
[S21max, i21] = max(S21dB);     f_S21max = f(i21);
[S11min, i11] = min(S11dB);     f_S11min = f(i11);

IL_at_S21max  = -S21max;                % IL at max S21
RL_at_S11min  = -S11min;                % RL at min S11

fprintf('\n================ PART A : VNA-STYLE RESULTS
================\n');
fprintf('Frequency points         : %d  (%.2f to %.2f GHz, step %.0f MHz)\n',...
      N, fstart/1e9, fstop/1e9, df/1e6);
fprintf('S21 maximum              : %.4f dB at %.3f GHz\n', S21max, f_S21max/1e9);
fprintf('Insertion loss at that f : %.4f dB\n', IL_at_S21max);
fprintf('S11 minimum              : %.4f dB at %.3f GHz\n', S11min, f_S11min/1e9);
fprintf('Return loss at that f    : %.4f dB\n', RL_at_S11min);

%% 5. PART B : ACCEPTABLE OPERATING REGION
```

```
ok = (S11dB <= S11_th) & (S21dB >= S21_th) & (RL >= 15) & (IL <= 3);

% Take the contiguous acceptable band that contains the S21-maximum point
if ~ok(i21)
    [~, ref] = min(abs(f - f0));
else
    ref = i21;
end
if any(ok)
    lo = ref; hi = ref;
    if ~ok(ref)       % fall back to the first acceptable point
        ref = find(ok,1,'first'); lo = ref; hi = ref;
    end
    while lo > 1 && ok(lo-1), lo = lo-1; end
    while hi < N && ok(hi+1), hi = hi+1; end
    fL = f(lo); fH = f(hi);
    fC = (fL + fH)/2;
    BW = fH - fL;
    FBW_meas = BW/fC*100;
else
    fL = NaN; fH = NaN; fC = NaN; BW = NaN; FBW_meas = NaN;
end

% Values at optimum frequency (taken at centre of acceptable region)
[~, ic] = min(abs(f - fC));
fopt      = f(ic);
S11_opt   = S11dB(ic);   S21_opt = S21dB(ic);
RL_opt    = RL(ic);      IL_opt  = IL(ic);
ISO_opt   = ISO(ic);
```

```
fprintf('\n============ PART B : NETWORK INTERPRETATION
==============\n');
fprintf('Lower frequency  fL      : %.3f GHz\n', fL/1e9);
fprintf('Upper frequency  fH      : %.3f GHz\n', fH/1e9);
fprintf('Centre frequency fC      : %.3f GHz\n', fC/1e9);
fprintf('Bandwidth  BW            : %.3f GHz  (%.2f %% fractional)\n', BW/1e9, FBW_meas);

%% 6. INPUT / OUTPUT MATCHING AND ISOLATION CHECKS
% Matching comparison S11 vs S22 inside the acceptable band
band = lo:hi;
fprintf('\nMean S11 in band         : %.2f dB\n', mean(S11dB(band)));
fprintf('Mean S22 in band         : %.2f dB\n', mean(S22dB(band)));
fprintf('Worst S11 in band        : %.2f dB\n', max(S11dB(band)));
fprintf('Worst S22 in band        : %.2f dB\n', max(S22dB(band)));
fprintf('Max |S12 - S21| (dB)     : %.3e   (reciprocity check)\n', ...
       max(abs(S12dB - S21dB)));
fprintf('Isolation at S21max freq : %.3f dB\n', ISO(i21));

% Part-3 style: RL, IL, ISO at S21 max
fprintf('\n--- At S21 maximum (%.3f GHz) ---\n', f_S21max/1e9);
fprintf('S11 = %.3f dB | S22 = %.3f dB | S21 = %.3f dB | S12 = %.3f dB\n', ...
   S11dB(i21), S22dB(i21), S21dB(i21), S12dB(i21));
fprintf('RL  = %.3f dB | IL = %.4f dB | Isolation = %.3f dB\n', ...
   RL(i21), IL(i21), ISO(i21));
% Part-4 style: values at S11 minimum
fprintf('--- At S11 minimum (%.3f GHz) ---\n', f_S11min/1e9);
fprintf('S11 = %.3f dB | S22 = %.3f dB | S21 = %.3f dB | S12 = %.3f dB\n', ...
   S11dB(i11), S22dB(i11), S21dB(i11), S12dB(i11));
```

```
fprintf('RL  = %.3f dB | IL = %.4f dB | Isolation = %.3f dB\n', ...
    RL(i11), IL(i11), ISO(i11));

%% 7. OUTPUT PARAMETER TABLE (Table D) WITH % DEVIATION
% Theoretical / expected values = specification limits (design targets)
th_S11 = -15;  th_S21 = -3;  th_RL = 15;  th_IL = 3;  th_f = f0/1e9;
pr_S11 = S11_opt; pr_S21 = S21_opt; pr_RL = RL_opt; pr_IL = IL_opt;
pr_f   = fopt/1e9;

dev = @(pr,th) abs(pr - th)/abs(th)*100;

T = table( ...
    {'S11 at fopt (dB)'; 'S21 at fopt (dB)'; 'Return loss (dB)'; ...
     'Insertion loss (dB)'; 'Optimum frequency (GHz)'}, ...
    [th_S11; th_S21; th_RL; th_IL; th_f], ...
    [pr_S11; pr_S21; pr_RL; pr_IL; pr_f], ...
    [dev(pr_S11,th_S11); dev(pr_S21,th_S21); dev(pr_RL,th_RL); ...
     dev(pr_IL,th_IL);  dev(pr_f,th_f)], ...
    'VariableNames', {'Parameter','Theoretical','Simulated','PercentDeviation'});
fprintf('\n================ OUTPUT PARAMETER TABLE
====================\n');
disp(T);

fprintf('Spec check at fopt  : S11<=-15 dB -> %s | S21>=-3 dB -> %s\n', ...
    ternary(S11_opt<=S11_th,'PASS','FAIL'), ternary(S21_opt>=S21_th,'PASS','FAIL'));
fprintf('Spec check at fopt  : RL>=15 dB   -> %s | IL<=3 dB   -> %s\n', ...
    ternary(RL_opt>=15,'PASS','FAIL'), ternary(IL_opt<=3,'PASS','FAIL'));

%% 8. PLOTS (VNA-style)
```

```
fG = f/1e9;
col = lines(4);

% (a) All four S-parameters
figure('Name','S-Parameters','Color','w','Position',[80 80 900 560]);
plot(fG,S11dB,'Color',col(1,:),'LineWidth',1.8); hold on;
plot(fG,S22dB,'--','Color',col(2,:),'LineWidth',1.8);
plot(fG,S21dB,'Color',col(3,:),'LineWidth',1.8);
plot(fG,S12dB,':','Color',col(4,:),'LineWidth',2.2);
yline(S11_th,'k--','S_{11} = -15 dB','LabelHorizontalAlignment','left');
yline(S21_th,'r--','S_{21} = -3 dB','LabelHorizontalAlignment','left');
yl = [-60 5]; ylim(yl);
patch([fL fH fH fL]/1e9, [yl(1) yl(1) yl(2) yl(2)], 'g', ...
    'FaceAlpha',0.12,'EdgeColor','none');
plot(f_S21max/1e9,S21max,'ko','MarkerFaceColor','y','MarkerSize',8);
plot(f_S11min/1e9,S11min,'ks','MarkerFaceColor','c','MarkerSize',8);
grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
title('Simulated S-Parameters of 50 \Omega Two-Port Network (8-12 GHz)');
legend('S_{11}','S_{22}','S_{21}','S_{12}','','','Acceptable region', ...
    'S_{21} max','S_{11} min','Location','south');
text(fC/1e9, -50, sprintf('f_C = %.2f GHz, BW = %.2f GHz',fC/1e9,BW/1e9), ...
    'HorizontalAlignment','center','FontWeight','bold');

% (b) Return loss and insertion loss
figure('Name','RL and IL','Color','w','Position',[100 100 900 520]);
yyaxis left;  plot(fG,RL,'LineWidth',1.8); ylabel('Return Loss (dB)');
yline(15,'--','RL = 15 dB');
yyaxis right; plot(fG,IL,'LineWidth',1.8); ylabel('Insertion Loss (dB)');
ylim([0 6]);
```

```
xlabel('Frequency (GHz)'); grid on;
title('Return Loss and Insertion Loss vs Frequency');
xline(fL/1e9,'k:'); xline(fH/1e9,'k:');

% (c) Zoomed passband view with limits
figure('Name','Passband zoom','Color','w','Position',[120 120 900 520]);
plot(fG,S21dB,'LineWidth',1.8); hold on;
plot(fG,S11dB,'LineWidth',1.8);
yline(S21_th,'r--'); yline(S11_th,'k--');
xlim([fL/1e9-0.8, fH/1e9+0.8]); ylim([-40 2]);
xline(fL/1e9,'g-','f_L'); xline(fH/1e9,'g-','f_H'); xline(fC/1e9,'m--','f_C');
grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
legend('S_{21}','S_{11}','Location','south');
title('Passband View: Acceptable Operating Region');

% (d) Reverse isolation and reciprocity
figure('Name','Isolation and reciprocity','Color','w','Position',[140 140 900 520]);
subplot(2,1,1);
plot(fG,ISO,'LineWidth',1.8); grid on;
ylabel('Isolation (dB)'); title('Reverse Isolation = -20log_{10}|S_{12}|');
subplot(2,1,2);
plot(fG,abs(S12dB-S21dB),'LineWidth',1.8); grid on;
xlabel('Frequency (GHz)'); ylabel('|S_{12}-S_{21}| (dB)');
title('Reciprocity Check (S_{12} \approx S_{21})');

% (e) Smith chart style polar plot of S11 and S22
figure('Name','Reflection on complex plane','Color','w');
plot(real(S11),imag(S11),'LineWidth',1.5); hold on;
plot(real(S22),imag(S22),'--','LineWidth',1.5);
```

```
th = linspace(0,2*pi,200); plot(cos(th),sin(th),'k');
axis equal; grid on; xlabel('Re(\Gamma)'); ylabel('Im(\Gamma)');
legend('S_{11}','S_{22}','|\Gamma| = 1'); title('Reflection Coefficients (Complex Plane)');

%% 9. EXPORT S-PARAMETER DATA (Touchstone-like CSV)
data = [f(:)/1e9, S11dB(:), S21dB(:), S12dB(:), S22dB(:), RL(:), IL(:), ISO(:)];
writematrix(data, 'Exp13_S_parameter_data.csv');
fprintf('\nData saved to Exp13_S_parameter_data.csv\n');

%% 10. FINAL CONCLUSION
fprintf('\n====================== CONCLUSION
==========================\n');
if any(ok)
    fprintf(['The 50 ohm two-port network meets S11 <= -15 dB and S21 >= -3 dB\n' ...
        'from %.2f GHz to %.2f GHz (f_C = %.2f GHz, BW = %.2f GHz).\n' ...
        'S12 ~ S21 confirms reciprocity and S11/S22 confirm matched ports.\n'], ...
        fL/1e9, fH/1e9, fC/1e9, BW/1e9);
else
    fprintf('The network does not satisfy the specified thresholds.\n');
end

%% Local helper function (must be at end of script in R2016b+)
function out = ternary(cond, a, b)
    if cond, out = a; else, out = b; end
end
```

Figure 1: S-Parameters

FIGURE FORMAT TOOLS LAYOUT

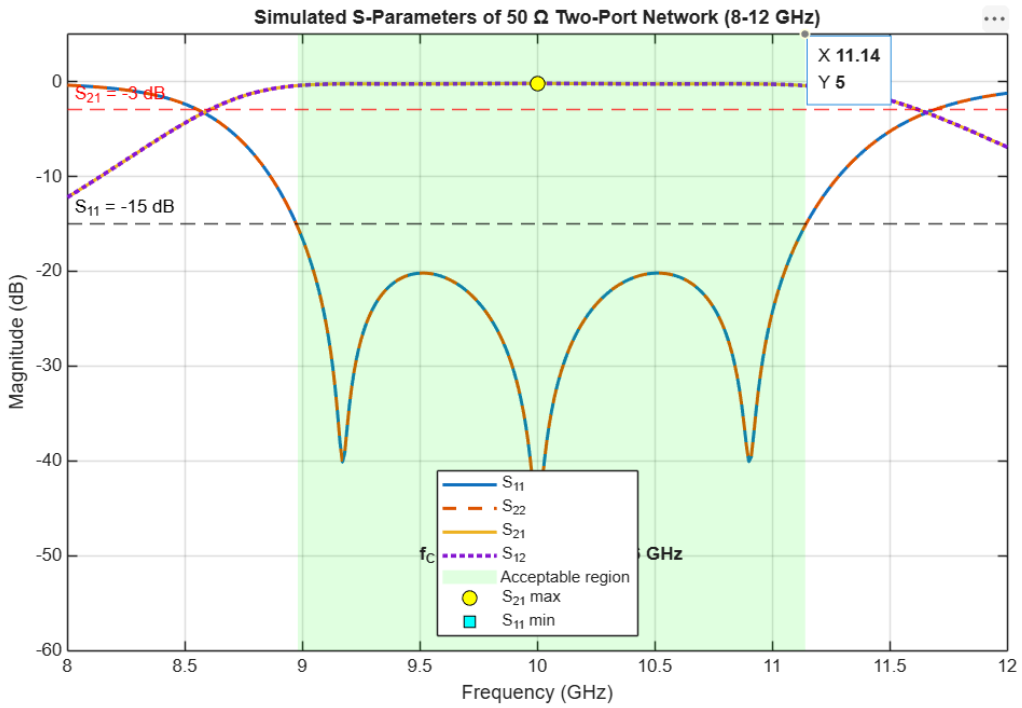


Figure 2: RL and IL

Figure 2: RL and IL FORMAT TOOLS LAYOUT

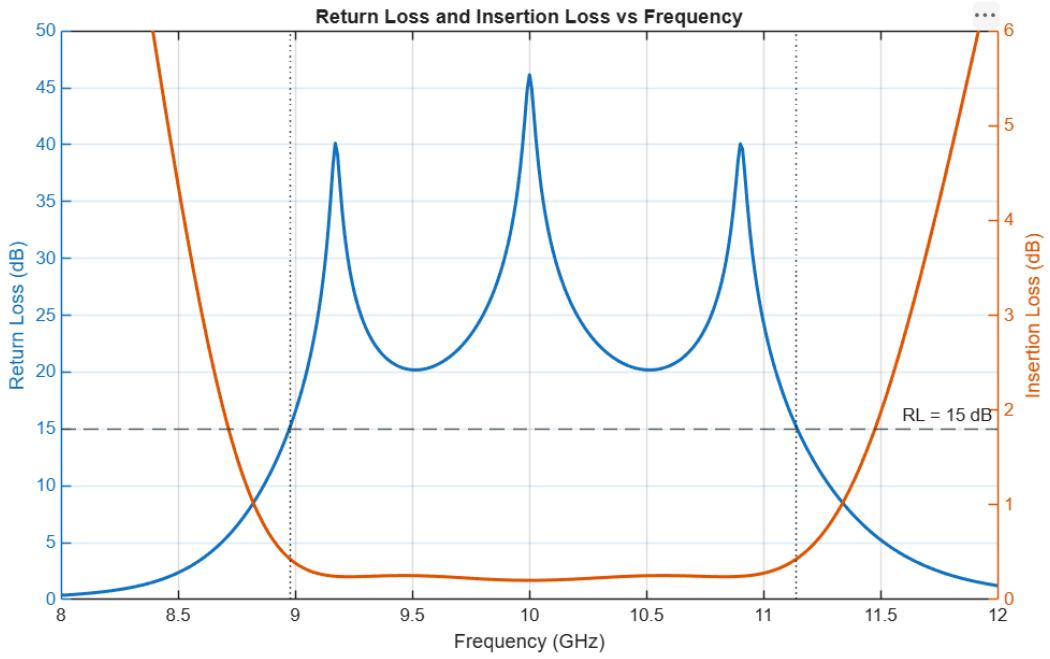


Figure 3: Passband zoom

FIGURE FORMAT TOOLS LAYOUT

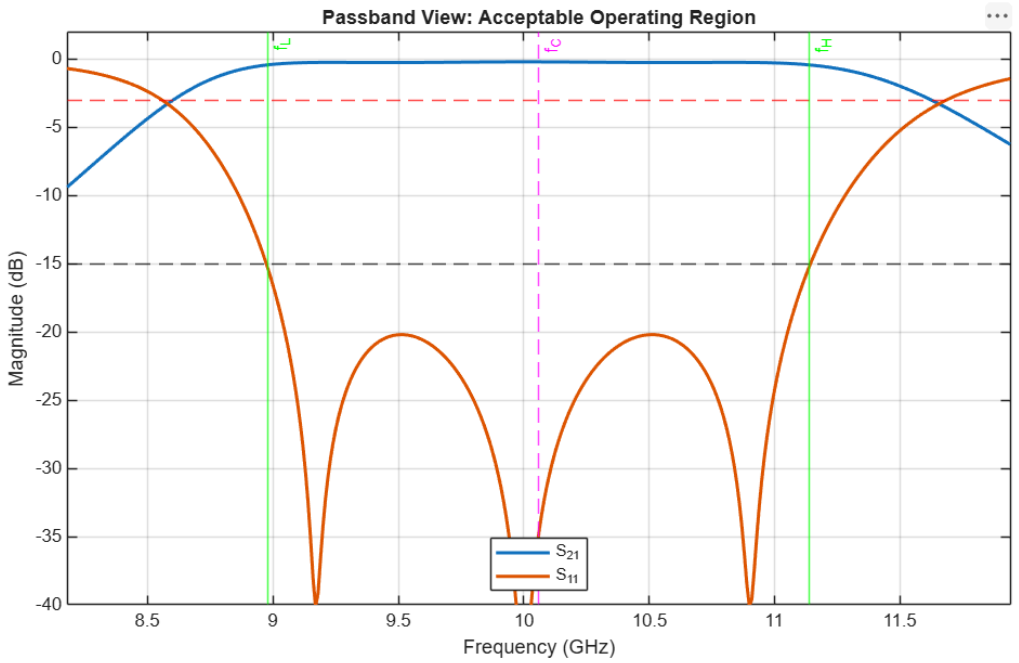
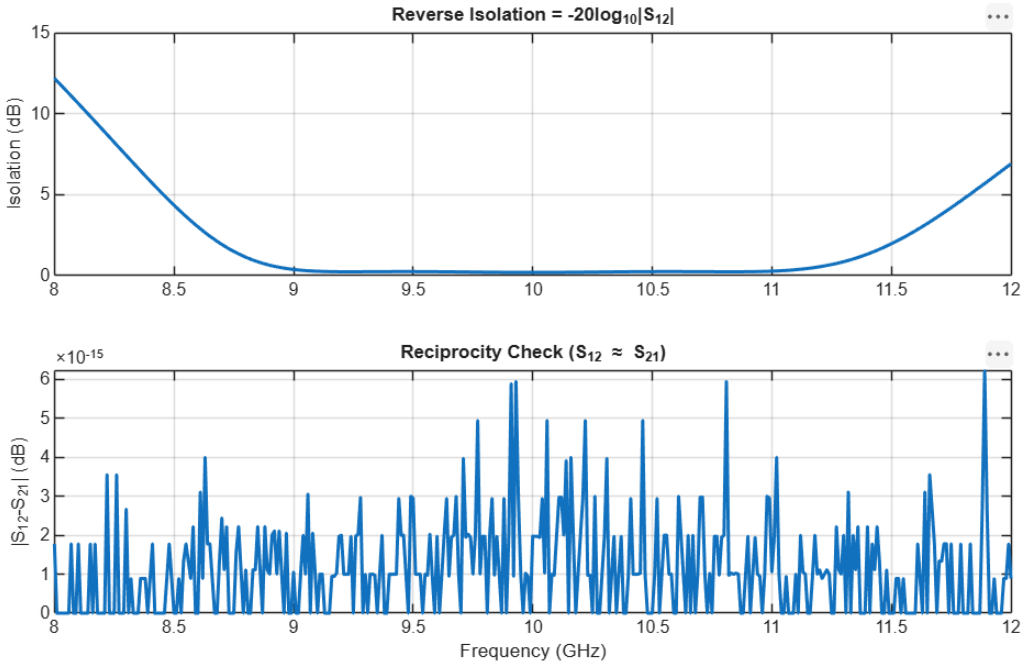
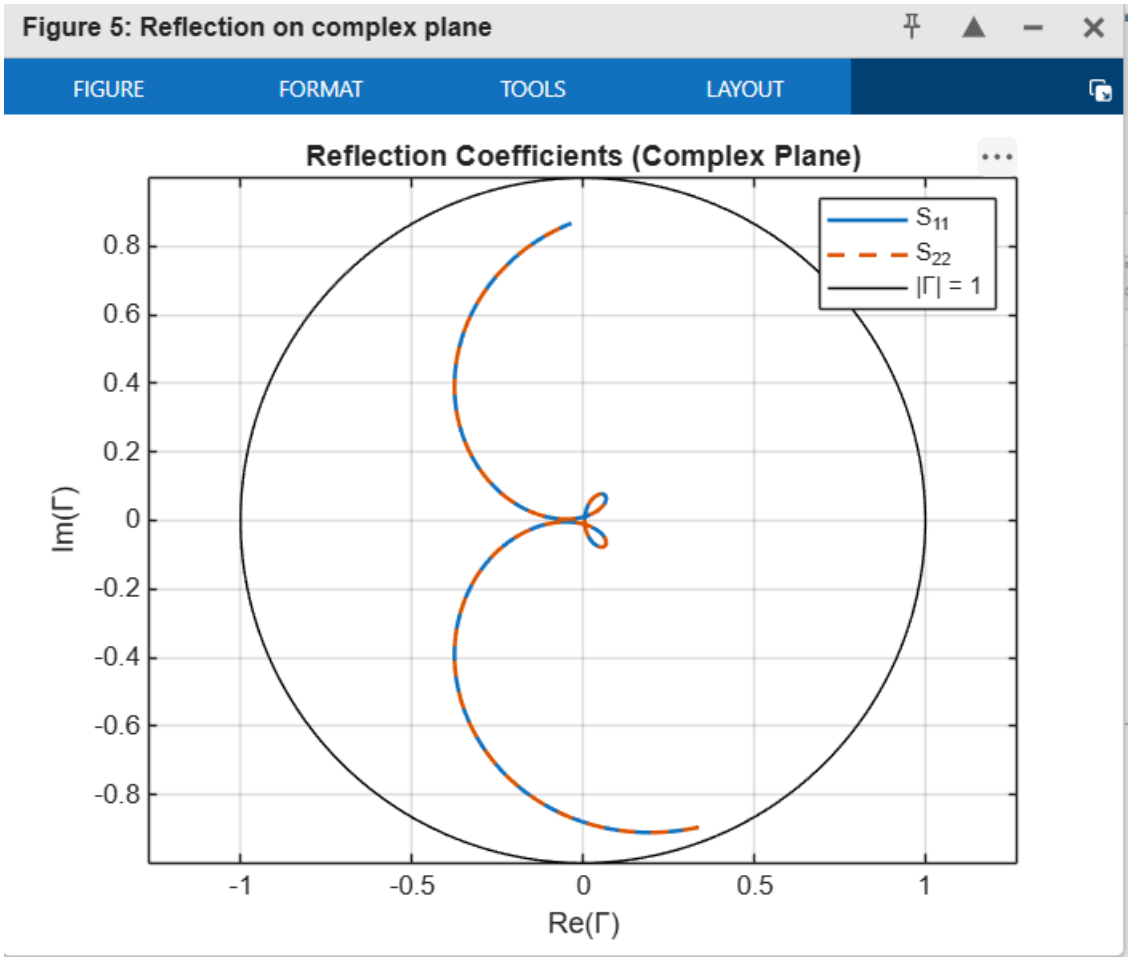


Figure 4: Isolation and reciprocity

FIGURE FORMAT TOOLS LAYOUT





Part A: VNA-style results

Quantity	**Value**
S21 maximum	−0.203 dB at 10.00 GHz
Insertion loss at that frequency	0.203 dB
S11 minimum	−46.18 dB at 10.00 GHz
Return loss at that frequency	46.18 dB

S21 and S11 reach their best values at the same frequency, 10.00 GHz.

Part B: acceptable operating region

Quantity	**Value**
Lower frequency fL	8.98 GHz
Upper frequency fH	11.14 GHz
Centre frequency fC = (fL + fH)/2	10.06 GHz

Quantity	Value
Bandwidth fH − fL	2.16 GHz (about 21.5% fractional)
Acceptable points	217 of 401, one continuous band

At the band edges, S11 is about −15.2 to −15.4 dB, which is why the S11 ≤ −15 dB limit sets the band. S21 is only about −0.43 dB there, so the −3 dB limit is never the limiting factor.

Matching and isolation

- **Input and output matching:** S11 and S22 are identical because the network is symmetric. Both average −25.1 dB across the band, and the worst case is −15.17 dB, so both ports are well matched.
- **Reciprocity:** the largest difference between S12 and S21 is about 6×10^{-15} dB, so S12 ≈ S21 holds.
- **Reverse isolation:** isolation equals insertion loss ($-20\log_{10}|S12| = -20\log_{10}|S21|$) in this reciprocal network. It is 0.203 dB at 10 GHz. The network is a passive reciprocal filter, so it gives no reverse isolation beyond its insertion loss.

Output parameter table (at fopt = 10.06 GHz)

Parameter	Theoretical (spec)	Simulated	% Deviation
S11 at fopt	≤ −15 dB	−34.79 dB	131.97
S21 at fopt	≥ −3 dB	−0.205 dB	93.18
Return loss	≥ 15 dB	34.79 dB	131.97
Insertion loss	≤ 3 dB	0.205 dB	93.18
Optimum frequency	10 GHz (design centre)	10.06 GHz	0.60

The theoretical column holds the specification limits, so the large deviations just show how much margin the network has beyond the limits. The simulated values all pass.

Conclusion

The 50 Ω two-port network meets S11 ≤ −15 dB and S21 ≥ −3 dB, and therefore RL ≥ 15 dB and IL ≤ 3 dB, from 8.98 to 11.14 GHz. The centre frequency is 10.06 GHz and the bandwidth is 2.16 GHz. The best matching and transmission both occur at 10.00 GHz, and the network is reciprocal with well-matched ports.

I can also write this up as a formatted lab report with the figures and tables if you want it.